

The Irreducible Three-Body Congruence Correlation and the Annihilation Lattice on the $6N$ Skeleton

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June 7, 2026

Abstract

Part XX recast the two-centre correlation of prime constellations as a circular convolution of castellated indicators on the phase-space torus $\mathbb{T}^1$. Here we extend the phase-space integration to three twin centres and ask whether the three-body correlation factorises into pairwise interactions. It does not. We define the irreducible three-body factor $U(j_1, j_2) = R_3/(R_2(j_1)R_2(j_2)R_2(j_2 - j_1))$ and find $U \neq 1$ at *every* lag pair: a genuine irreducible three-body congruence term is generic, not exceptional. We then state the correct criterion for total annihilation ($R = 0$): the union of the constellations' dead-residue sets must *tile* $\mathbb{Z}/q$ for some prime q . This corrects a natural but false dimension-count heuristic ($m_A + m_B \geq q$): tiling requires the hole-sets to be *complementary*, not merely numerous. A triplet contains a twin, so their dead-sets always overlap and a Twin-Triplet pair never annihilates ($I_q \geq 1/q > 0$); the lowest order at which twin-based annihilation occurs is therefore *three*-body—three consecutive twin centres tile $\mathbb{Z}/5$, recovering the mod-5 inadmissibility of a six-member skeleton window. We map the resulting annihilation lattice and the finite irreducible term. Closed-form throughout; the integral equals the discrete singular-series ratio at integer lags; no infinitude is claimed.

1 Introduction

Part XX established that the spatial correlation of two twin centres is a circular convolution of castellated indicators on $\mathbb{T}^1$, quantised into three interference states [6]. A two-body correlation, however, cannot by itself reveal whether the lattice of twin centres behaves as a gas of independent pairs or carries genuine higher-order structure. The natural next question is whether the three-body correlation $R_3(j_1, j_2)$ of twins at $N, N + j_1, N + j_2$ factorises into the product of the three pairwise correlations of Part XX.

We show it does not, anywhere. The deviation defines an *irreducible three-body congruence term* that is generic across the lattice, and whose extreme limit—total annihilation, $R_3 = 0$ —obeys a precise tiling criterion that we state and that corrects a tempting dimension-count heuristic. As in Volume I, we separate axes carefully: the annihilation studied here is a *horizontal* (inter-centre) phenomenon, distinct from the single-centre “killer prime” that drives the *vertical* ω -collapse of Volume I [4, 7].

2 The three-body joint measure

For a prime $q > 3$ a twin centre N is forbidden modulo q on the two dead residues $\text{dead}(q) = \{6^{-1}, -6^{-1}\}$. A twin at $N + j$ has the shifted set $\text{dead}(q) - j \pmod{q}$. For twins at $N, N + j_1, N + j_2$ (taking $j_0 = 0$) the joint survival measure at q is the transmitting fraction of the union,

$$I_q(j_1, j_2) = \frac{q - \nu_q}{q}, \quad \nu_q = \left| \text{dead}(q) \cup (\text{dead}(q) - j_1) \cup (\text{dead}(q) - j_2) \right|, \quad (1)$$

and the normalised three-body correlation, with $\mathfrak{S}$ the singular series, is

$$R_3(j_1, j_2) = \frac{\mathfrak{S}(\text{all three})}{\mathfrak{S}_{\text{twin}}^3} = \prod_{q>3} \frac{1 - \nu_q/q}{(1 - 2/q)^3}. \quad (2)$$

The pairwise correlation of Part XX is $R_2(j) = \prod_q (1 - \nu_q^{(2)}/q)/(1 - 2/q)^2$ with $\nu_q^{(2)} = |\text{dead}(q) \cup (\text{dead}(q) - j)|$.

Definition 1 (Irreducible three-body factor).

$$U(j_1, j_2) = \frac{R_3(j_1, j_2)}{R_2(j_1) R_2(j_2) R_2(j_2 - j_1)}. \quad (3)$$

$U \equiv 1$ would mean the three-body correlation is fully explained by its three pairwise interactions. Any deviation is an irreducible three-body term.

By inclusion–exclusion, $\nu_q = \sum |A_i| - \sum |A_i \cap A_j| + |A_1 \cap A_2 \cap A_3|$; the three pairwise factors absorb the singletons and the pairwise intersections, so U is driven by the triple intersection $|A_1 \cap A_2 \cap A_3|$ —a residue killed by all three centres at once—together with the saturation that occurs when the union fills $\mathbb{Z}/q$.

3 The annihilation criterion: tiling, not counting

Theorem 1 (Annihilation is tiling). *The joint density of constellations $A_1, \dots, A_k$ at centres $N + j_1, \dots, N + j_k$ vanishes, $R = 0$, if and only if there exists a prime $q > 3$ at which the shifted dead-sets cover every residue,*

$$\bigcup_{i=1}^k (\text{dead}_{A_i}(q) - j_i) = \mathbb{Z}/q\mathbb{Z}.$$

Proof. $R = 0$ iff some local factor $1 - \nu_q/q = 0$, i.e. $\nu_q = q$, i.e. the union of the shifted dead-sets is all of $\mathbb{Z}/q$. A single vanishing prime annihilates the whole product. $\square$

This makes precise—and corrects—the intuition that “dimension sum $\geq q$ forces collapse.” Covering $\mathbb{Z}/q$ requires the dead-sets to *tile*: to be numerous *and* complementary. Mere counting is not enough.

Corollary 1 (No Twin–Triplet annihilation). *A triplet $(6N - 1, 6N + 1, 6N + 5)$ contains the twin $(6N - 1, 6N + 1)$, so at $q = 5$ $\text{dead}_{\text{trip}}(5) = \{0, 1, 4\} \supseteq \{1, 4\} = \text{dead}_{\text{twin}}(5)$. For every shift j the two sets overlap, $\text{dead}_{\text{twin}} \cap (\text{dead}_{\text{trip}} - j) \neq \emptyset$, so they can never tile $\mathbb{Z}/5$; direct evaluation gives $I_5(j) \in \{1/5, 2/5\}$, never 0. A Twin–Triplet pair does not annihilate at any lag; indeed no two-body pair built on a shared twin does. The twin’s two-hole budget is simply too small to complete a tiling against a partner that already contains it.*

The lowest order at which twin-based annihilation can occur is thus $k = 3$: three twins contribute $2k = 6 \geq 5$ holes, enough to tile the smallest prime $q = 5$.

4 The irreducible term and the annihilation lattice

Evaluating $U(j_1, j_2)$ over the lattice $1 \leq j_1, j_2 \leq 30$ (singular-series cutoff $q \leq 59$) gives the central empirical facts of this paper, shown in Fig. 1.

Three-body terms are generic. Across all 870 ordered lag pairs in the window, $U = 1$ occurs *nowhere*: the three-body correlation never reduces to its pairwise parts. The finite irreducible

Three-body twin correlation $U(j_1, j_2) = R_3/(R_2 R_2 R_2)$: irreducible everywhere, annihilating on a lattice

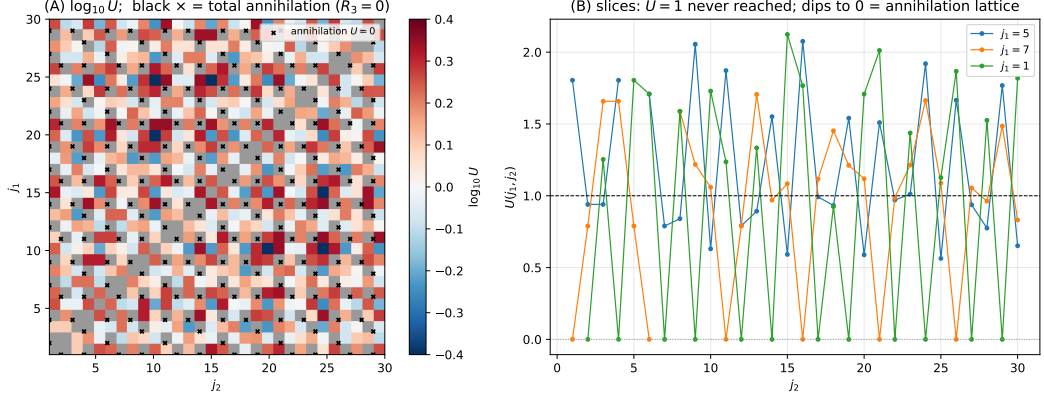


Figure 1: (A) $\log_{10} U(j_1, j_2)$ for three twin centres; black crosses mark the annihilation lattice $U = 0$ ($R_3 = 0$). (B) Slices $j_1 \in \{1, 5, 7\}$: $U = 1$ (independence of the three-body term) is reached nowhere, and the curve dips to 0 on the annihilation lattice.

term ranges from $U \approx 0.37$ to $U \approx 2.24$; its sharpest finite values arise from a triple coincidence, a prime at which all three centres share a dead residue (e.g. $j_1 = 5, j_2 = 10$, where $0, 5, 10$ are all $\equiv 0 \pmod{5}$) and the three twin holes align, $|A_1 \cap A_2 \cap A_3| = 2$ at $q = 5$).

The annihilation lattice. Of the scanned pairs, 216 give $U = 0$: a periodic lattice of forbidden configurations. The defining instance is $(j_1, j_2) = (1, 2)$ —three consecutive twin centres. At $q = 5$,

$$\{1, 4\} \cup (\{1, 4\} - 1) \cup (\{1, 4\} - 2) = \{1, 4\} \cup \{0, 3\} \cup \{2, 4\} = \mathbb{Z}/5,$$

so the six skeleton members $6N - 1, 6N + 1, 6N + 5, 6N + 7, 6N + 11, 6N + 13$ always contain a multiple of 5: three consecutive twin centres are impossible beyond $6N = 5$. Theorem 1 thus reproduces, from pure phase-space tiling, the mod-5 inadmissibility that classical k -tuple theory assigns to this six-member window. The annihilation lattice is the admissibility constraint seen as a congruence diffraction zero.

5 Twin immunity, restated

Lemma 1 (Two-body twin immunity). *No two-body correlation built on a twin annihilates: for any partner X , $R_2 = 0$ would require $\text{dead}_{\text{twin}}(q)$ and $\text{dead}_X(q) - j$ to tile $\mathbb{Z}/q$, impossible for $q \geq 5$ whenever the partner shares the twin’s members (Corollary 1). In particular Twin–Twin has $I_q \geq (q - 4)/q \geq 1/5 > 0$.*

We stress what this does *not* say. Two-body immunity is not unique to the twin, and it is not the source of the Volume I “killer prime” collapse. That collapse is a *single-centre* effect—a cross-centre tuple’s own member $6N + 5$ is divisible by 5 whenever $5 \mid N$, i.e. $0 \in \text{dead}(5)$ —and lives on the vertical ω -axis [4]. The horizontal annihilation of this paper is a distinct, higher-order phenomenon: it needs three bodies, and it encodes admissibility rather than divisibility of a single member.

6 Conclusion

Extending the phase-space convolution to three centres shows that the lattice of twin centres is not a gas of independent pairs: an irreducible three-body congruence term is present at every lag, and its zeros form a periodic annihilation lattice governed by a tiling criterion (Theorem 1). The

criterion corrects the naive dimension count—tiling demands complementarity, not abundance—and explains why a Twin–Triplet pair, despite $m_A + m_B = 5$, never annihilates, while three consecutive twins do, recovering mod-5 admissibility from geometry alone. The natural next step is the general k -body tiling spectrum and its relation to the Hardy–Littlewood admissibility of long windows on the $6N$ skeleton. As throughout the programme, the construction is closed-form, equal to the discrete singular-series ratio at integer lags, and makes no claim about the infinitude of any constellation.

Data and code availability. The three-body computation (the irreducible factor U , the annihilation count, and the figure) and a verifier of the tiling criterion are openly available [8].

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